Euclid's Elements — Book XI
Proposition 5

Formal Reconstruction — Technical Documentation

Jurek Różański
Military University of Technology, Warsaw, Poland
jerzy.rozanski@wat.edu.pl

CGJteamLab

If a straight line is set up at right angles to three straight lines which meet one another at their common point of section, then the three straight lines lie in one plane.

1 Introduction

Proposition XI.5 is the first Book XI proposition whose conclusion is a genuinely spatial coplanarity statement forced by perpendicularity.

A line l passes through a point O and is perpendicular there to three lines m, n, p. Two of these lines, m and n, are assumed distinct. The conclusion is that m, n, p lie in one plane.

The formal reconstruction is not a line-by-line transcription of Euclid's proof. The current spatial interface makes it possible to use Proposition XI.4 twice and thereby obtain a short contradiction argument.

2 Euclid's Statement

Euclid's Proposition XI.5 states:

> If a straight line is set up at right angles to three straight lines which meet one another at their common point of section, then the three straight lines lie in one plane.

In Euclid's notation the common perpendicular is AB, and the three lines through the common point B are BC, BD, and BE.

3 Classical Configuration

Euclid assumes, for contradiction, that two of the three lines, say BD and BE, lie in a reference plane while BC lies in a more elevated plane.

He then considers the plane through AB and BC. Its intersection with the reference plane is a line BF. Thus AB, BC, BF lie in one plane.

At the same time, since AB is perpendicular to BD and BE, Proposition XI.4 implies that AB is perpendicular to the reference plane. Therefore AB is also perpendicular to BF.

The contradiction is that, in the plane through AB and BC, the lines BC and BF would both form a right angle with AB at the same point.

4 Classical Proof

The local classical dependency pattern is:

$$\text{XI.3} \longrightarrow \text{intersection line } BF,$$

$$\text{XI.4} \longrightarrow AB \perp (\text{reference plane}),$$

followed by Euclid's Definition XI.3, which turns line–plane perpendicularity into perpendicularity to every line of the plane meeting the perpendicular.

Thus the classical proof is a contradiction proof based on the intersection of two planes and on the impossibility of two distinct perpendiculars to the same line at the same point in one plane.

5 What is genuinely three-dimensional here?

The essential spatial content is not the elementary fact about right angles. It is the management of planes.

The proof must establish:

- a plane through two intersecting lines;
- a second plane through another pair of intersecting lines;
- that these two planes are distinct;
- a line of intersection of the two planes;
- containment of relevant lines in the appropriate planes.

The current Lean proof does not open a new planar `PlaneGeo` argument of its own. Instead, its planar geometric content is inherited through the already proved Proposition XI.4.

6 Formal Statement

The production theorem is:

```
theorem euclid_proposition_11_5
    [H : HilbertIncidence Geo]
    [HilbertPlaneIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    [HSI : HilbertSpaceIncidence Geo]
    [_HSO : HilbertSpaceOrder
      (Geo := Geo) (H := H) (S := S)]
    [_HSC : HilbertSpaceCongruence
      (Geo := Geo) (H := H) (S := S)]
    (l m n p : Geo.Line)
    (O : Geo.Point)
    (hmn : Ne m n)
    (hPerpM :
      HilbertLinesPerpendicularAt Geo l m O)
    (hPerpN :
      HilbertLinesPerpendicularAt Geo l n O)
    (hPerpP :
      HilbertLinesPerpendicularAt Geo l p O) :
    exists pi : S.Plane,
      HilbertLineInPlane Geo m pi /\
      HilbertLineInPlane Geo n pi /\
      HilbertLineInPlane Geo p pi
```

The conclusion is constructive at the level of incidence: Lean returns an explicit plane π containing all three lines.

The hypothesis `hmn : Ne m n` makes explicit a point that is usually left implicit in the classical diagram. The two intersecting lines m, n must be distinct in order to determine the reference plane.

7 Spatial / Planar Architecture Used

The proposition uses the ambient three-dimensional layer:

- `HilbertSpacePrimitive` for planes and point–plane incidence;
- `HilbertSpaceIncidence` for spatial plane constructions;
- `HilbertLineInPlane` for line–plane containment;
- `HilbertLinesPerpendicularAt` for spatial line–line perpendicularity;
- `HilbertLinePerpendicularPlaneAt` for line–plane perpendicularity.

The ambient geometry does not install the planar order and congruence classes as global instances on the spatial object. The architectural distinction is:

```
-- not global ambient instances:
```

```
HilbertOrder Geo
HilbertCongruence Geo

-- spatial layers:
HilbertSpaceOrder
HilbertSpaceCongruence
```

Planar order and congruence enter only through the established `PlaneGeo` machinery used inside Proposition XI.4.

8 Proof Architecture of the Reconstruction

Let m and n be the first two lines perpendicular to l at O.

8.1 Step 1: construct the reference plane

Since $m \neq n$ and both pass through O, the theorem

```
hilbert_plane_through_two_intersecting_lines
```

constructs a plane π containing m and n.

8.2 Step 2: apply Proposition XI.4

The lines m and n are packaged as `PlaneLine`s of π, and O as a `PlanePoint` of π.

Proposition XI.4 then yields

$$l \perp \pi \quad \text{at } O.$$

In Lean this is the call to

```
euclid_proposition_11_4
```

with the two given perpendicularities $l \perp m$ and $l \perp n$.

8.3 Step 3: split on whether p already lies in π

If

$$p \subset \pi,$$

the theorem is finished immediately.

Assume therefore

$$p \not\subset \pi.$$

Since $l \perp p$, the lines l and p are distinct. They meet at O, so they determine another plane σ.

8.4 Step 4: intersect the two planes

The planes π and σ must be distinct. If $\pi = \sigma$, then $p \subset \sigma$ would imply $p \subset \pi$, contrary to the case assumption.

The theorem

```
hilbert_plane_intersection_line
```

therefore gives a line q through O contained in both planes:

$$q \subset \pi, \qquad q \subset \sigma.$$

Since $l \perp \pi$ and q is a line of π through O,

$$l \perp q.$$

This is obtained directly from

```
HilbertLinePerpendicularPlaneAt.perpendicular_to_line
```

8.5 Step 5: the second use of XI.4

Now p and q both lie in σ, and

$$l \perp p, \qquad l \perp q.$$

Suppose $p \neq q$. Applying Proposition XI.4 inside σ gives

$$l \perp \sigma.$$

But $l \subset \sigma$ by construction. This is impossible.

Hence

$$p = q.$$

Since $q \subset \pi$, it follows that $p \subset \pi$, and therefore

$$m, n, p \subset \pi.$$

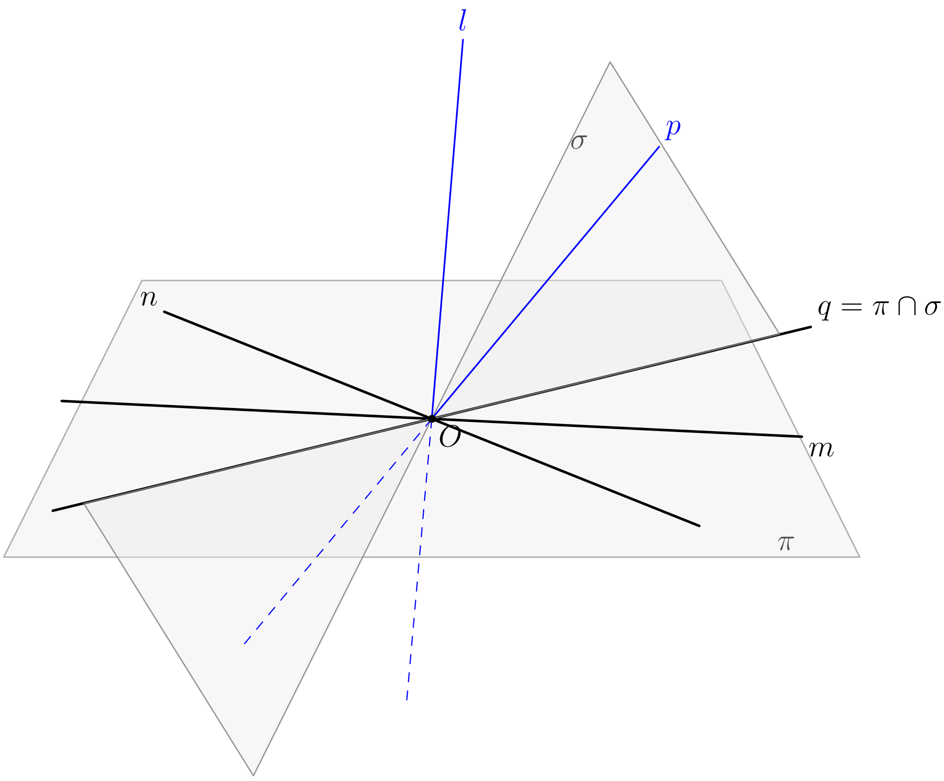

Figure 1: The formal contradiction argument for Proposition XI.5. The black lines m, n, q lie in the reference plane π. The spatial lines l, p are blue. Under the temporary assumption $p \not\subset \pi$, the lines l, p determine the second plane σ, and $q = \pi \cap \sigma$. If $p \neq q$, a second application of XI.4 would make l perpendicular to σ, although $l \subset \sigma$.

9 Lean Formalization

The small proposition-local helper is:

```
theorem hilbert_linePerpendicularPlaneAt_not_in_plane
    [H : HilbertIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    (l : Geo.Line)
    (pi : S.Plane)
    (O : Geo.Point)
    (hPerp :
      HilbertLinePerpendicularPlaneAt Geo l pi O) :
    Not (HilbertLineInPlane Geo l pi) := by

  intro hlpi

  have hSelf :
      HilbertLinesPerpendicularAt Geo l l O :=
    HilbertLinePerpendicularPlaneAt.perpendicular_to_line
      (Geo := Geo)
      hPerp
      hlpi
      hPerp.1

  exact
    (hilbert_linesPerpendicularAt_ne
      Geo l l O hSelf) rfl
```

This helper is purely definitional in geometric content. If a line perpendicular to a plane were itself contained in the plane, then the universal line–plane perpendicularity condition would make the line perpendicular to itself. The predicate `HilbertLinesPerpendicularAt` is nondegenerate, so this is impossible.

The theorem body then consists of two plane constructions, one plane-intersection construction, and two calls to XI.4.

10 Hidden Formal Data

Several facts visible in a classical spatial diagram must be supplied explicitly in Lean.

- O lies in the plane determined by m, n.
- The ambient lines m, n must be packaged as `PlaneLine Geo pi`.
- Their distinctness must be transported to the subtype level.
- If $p \not\subset \pi$, the plane through l, p must be constructed explicitly.
- The two planes must be proved distinct before their intersection line can be used.
- The intersection line q must carry proofs that it lies in both π and σ.
- For the second XI.4 call, p, q and O are repackaged as `PlaneLine` and `PlanePoint` objects of σ.

These are mathematical incidence obligations, not merely syntactic casts. They encode information that Euclid's diagram leaves implicit.

11 Relationship with Earlier Book XI Propositions

The classical proof explicitly uses XI.3 and XI.4.

The Lean proof imports Proposition XI.4 and uses the current spatial intersection theorem

```
hilbert_plane_intersection_line
```

instead of importing a separate formal Proposition XI.3.

Thus the two dependency graphs are not identical.

The actual Lean Book XI dependency is:

$$\boxed{\text{XI.4}} \longrightarrow \boxed{\text{XI.5}}.$$

Proposition XI.5 is then the coplanarity tool required by the classical route to Proposition XI.6.

12 Relationship with Book I / Book Zero / Hilbert Infrastructure

No Book I proposition is called directly by the XI.5 theorem body.

The inherited planar Hilbert and Book Zero infrastructure enters transitively through Proposition XI.4. In particular, XI.4 internally uses induced plane geometry, while XI.5 itself is primarily a spatial incidence argument.

The new helper `hilbert_linePerpendicularPlaneAt_not_in_plane` does not add a geometric axiom; it follows from the existing definitions and nondegeneracy of line–line perpendicularity.

13 Relationship with the Foundations

XI.5 is neutral in the current reconstruction. Its proof uses spatial incidence, order, congruence, Proposition XI.4, and explicit plane intersection arguments, but no Hilbert Group IV assumption and no continuity principle.

The important foundational point is therefore architectural rather than axiomatic: coplanarity is never inferred from the picture. The proof must construct or identify the relevant plane and carry the corresponding line-in-plane witnesses explicitly.

14 Dependency Summary

The actual formal dependency chain is:

Hilbert incidence / order / congruence infrastructure
$\downarrow$
spatial primitive / incidence / order / congruence
$\downarrow$
PlaneGeo and plane–space bridges used by XI.4
$\downarrow$
`euclid_proposition_11_4`
$\downarrow$
plane through intersecting lines
intersection line of two planes
`hilbert_linePerpendicularPlaneAt_not_in_plane`
$\downarrow$
`euclid_proposition_11_5`.

At the current checkpoint the production theorem compiles locally and its axiom audit has been run successfully. A full repository `lake build` is a separate checkpoint and should be recorded only after it is run.

15 Conclusion

Proposition XI.5 identifies the plane perpendicular to a given line at a given point as the unique carrier of all lines through that point perpendicular to the line, in the precise three-line form stated by Euclid.

The formal proof is shorter than the classical local proof graph because the present 3D interface already contains an intersection-line theorem for two planes and because XI.4 can be applied a second time to exclude the hypothetical second plane.

No coordinates, vectors, scalar products, continuity principle, or new spatial axiom enter the proof.